An Introduction to Sex Ratio Theory

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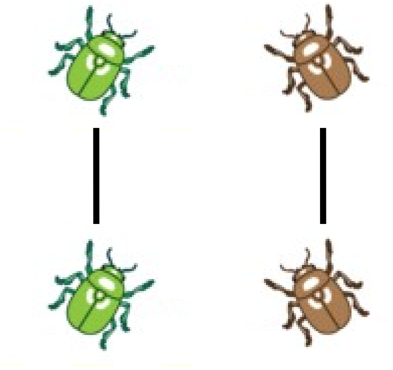
Evolution by natural selection

Evolution:
change in allele frequencies in a population over time

Three conditions for evolution to happen (Lewontin, 1970):

(1) **phenotypic variation**

(2) **heredity** (offspring resemble their parents)



(3) **fitness differences**

What is fitness in evolutionary biology?

Fitness:
number of surviving offspring

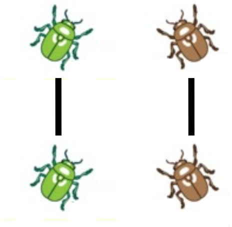
$w_{green} = 2$ $w_{brown} = 4$

(in mathematical models, fitness is often denoted by *w*)

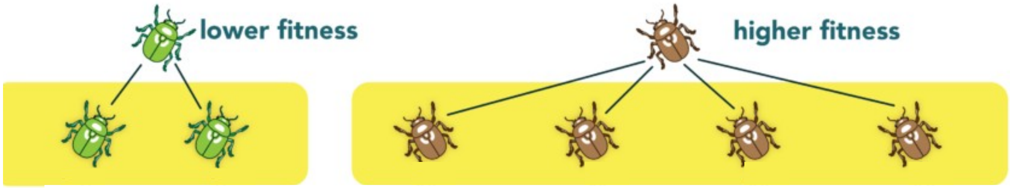
Evolution by natural selection

Lewontin’s three conditions:

(1) **phenotypic variation** 

(2) **heredity** (offspring resemble their parents) 

(3) **fitness differences**



If all three conditions are fulfilled, evolution by natural selection can happen!

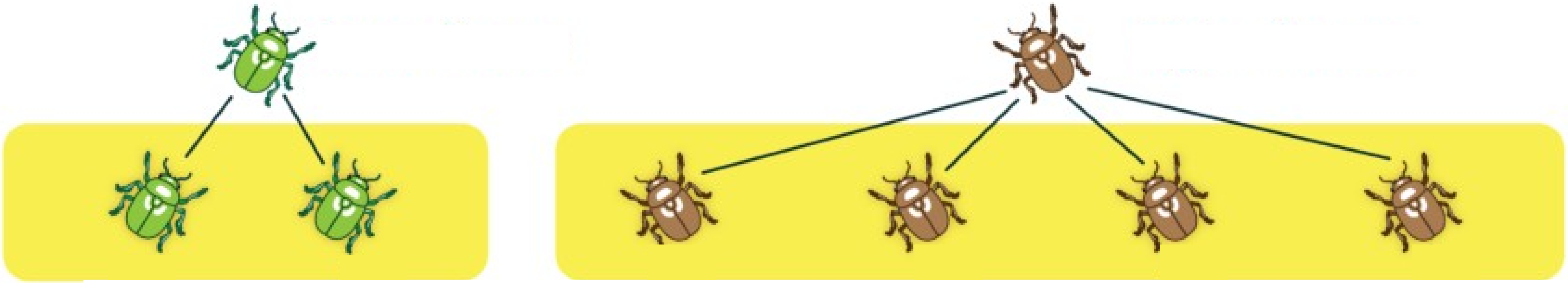
Question: How do genotype frequencies change over time?

A very simple model of evolution by natural selection

Population size	$N = 2$
Genotypes	$n = 1, 2$
Frequencies	$x_1 = x_2 = 0.5$
Fitness	$w_1 = 2$, $w_2 = 4$

Parent generation

Offspring generation



Number of offspring:

$x_1 N w_1 = 0.5 \cdot 2 \cdot 2 = 2$

$x_2 N w_2 = 0.5 \cdot 2 \cdot 4 = 4$

New frequencies:

$x_1' = x_1 N w_1 / (x_1 N w_1 + x_2 N w_2)$
$= 2 / (2 + 4) = 1/3$

$x_2' = x_2 N w_2 / (x_1 N w_1 + x_2 N w_2)$
$= 4 / (2 + 4) = 2/3$

Generalizing the simple model

Frequency of genotype *i* in the next generation:

Population size	N
Genotypes	$1, \ldots, n$
Frequencies	$x_1, \ldots, x_n$
Fitness	$w_1, \ldots, w_n$

$$x_i' = \frac{\overbrace{x_i N w_i}^{\text{number of offspring of genotype } i}}{\underbrace{\sum_{j=1}^{n} x_j N w_j}_{\text{number of offspring of all genotypes}}} = \frac{x_i w_i}{\sum_{j=1}^{n} x_j w_j}$$

Average fitness of the population: $W = \sum_{j=1}^{n} x_j w_j$

$$\boxed{x_i' = x_i \frac{w_i}{W}}$$

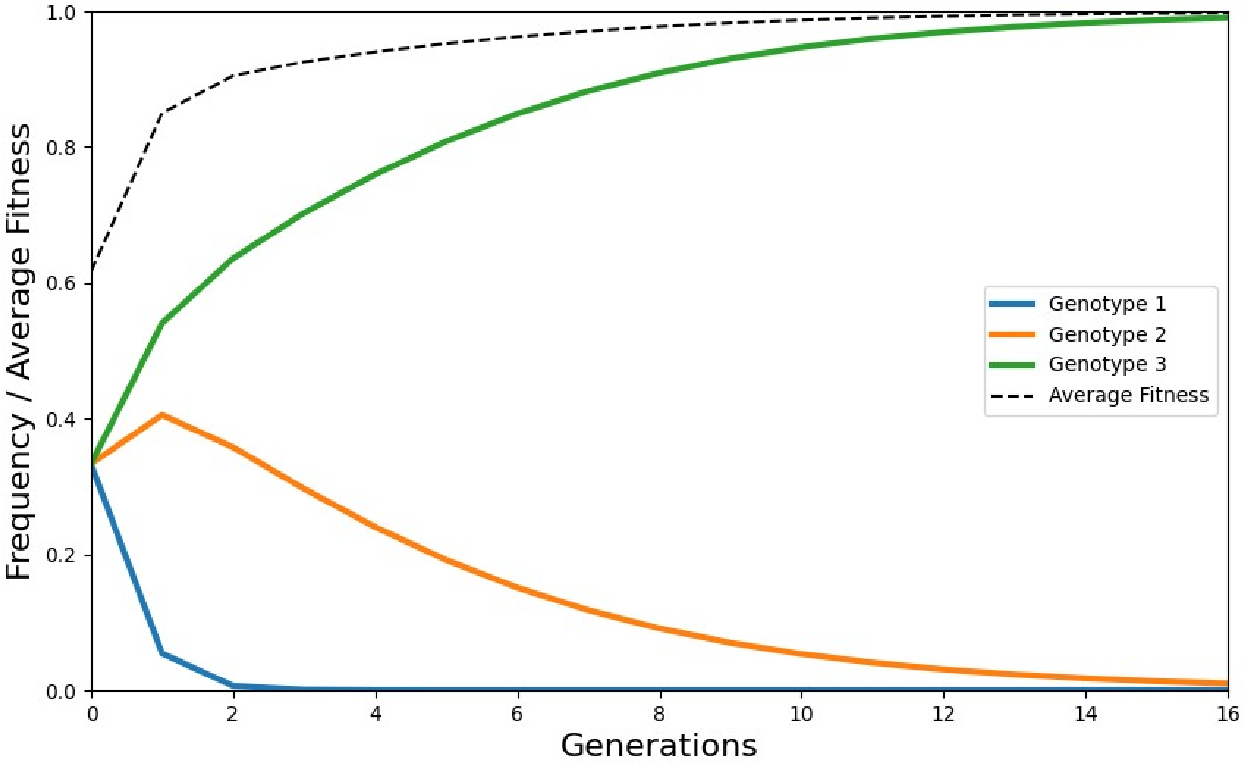
Replicator equation

What do we learn from the replicator equation?

$x_1 = x_2 = x_3 = 1/3$

$w_1 = 0.1$
$w_2 = 0.75$
$w_3 = 1$

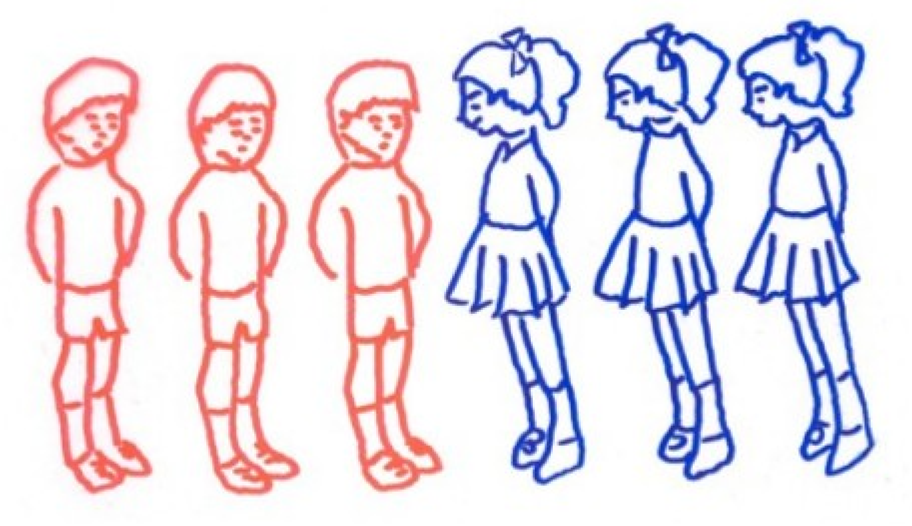
$$\boxed{x_i' = x_i \frac{w_i}{W}}$$



A phenotype increases in frequency if its fitness is greater than the average fitness of the population

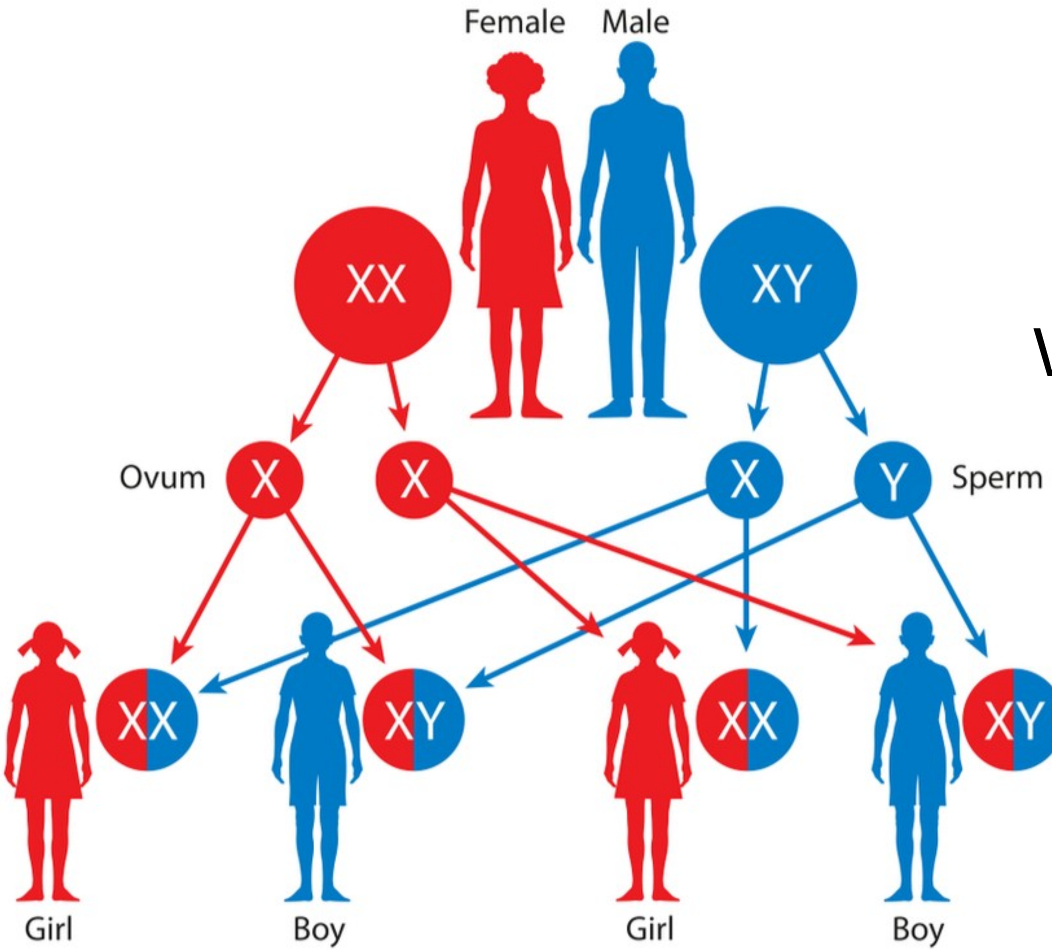
Sex ratio theory

Why is the offspring sex ratio so balanced
in humans and many other animal species?



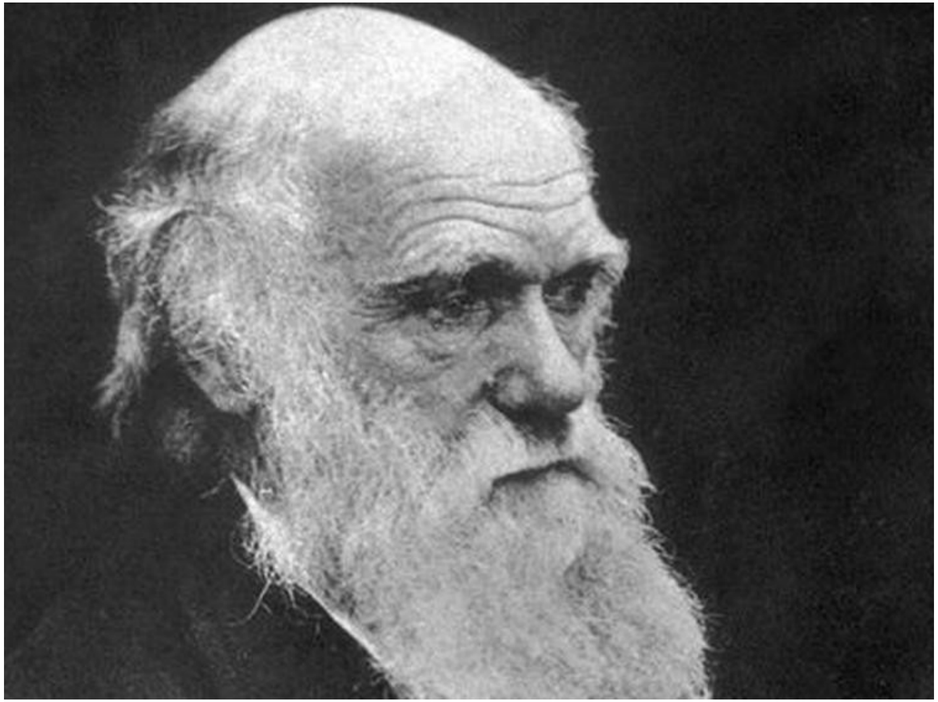
Explaining balanced sex ratios...

But that's a **proximate** explanation! (How?)



We're interested in an **ultimate** explanation! (Why?)

Even Darwin was puzzled by even sex ratios...



“I formerly thought that when a tendency to produce the two sexes in equal numbers was advantageous [...], it would follow from natural selection, but I now see that the whole problem is so intricate that it is safer to leave its solution for the future.”

(Darwin, *The Descent of Man*, 1874)

So, again: Why is there a 50:50 sex ratio in so many species?

A very simple thought experiment

Female 1:	3 sons, 1 daughter	♂ ♂ ♂ ♀
Female 2:	2 sons, 2 daughters	♂ ♂ ♀ ♀
Female 3:	1 son, 3 daughters	♂ ♀ ♀ ♀

If sons and daughters have the same probability of survival,
all these females are equally successful in terms of (simple) fitness

The sex ratio does not seem to matter at all!

A slightly more complex thought experiment

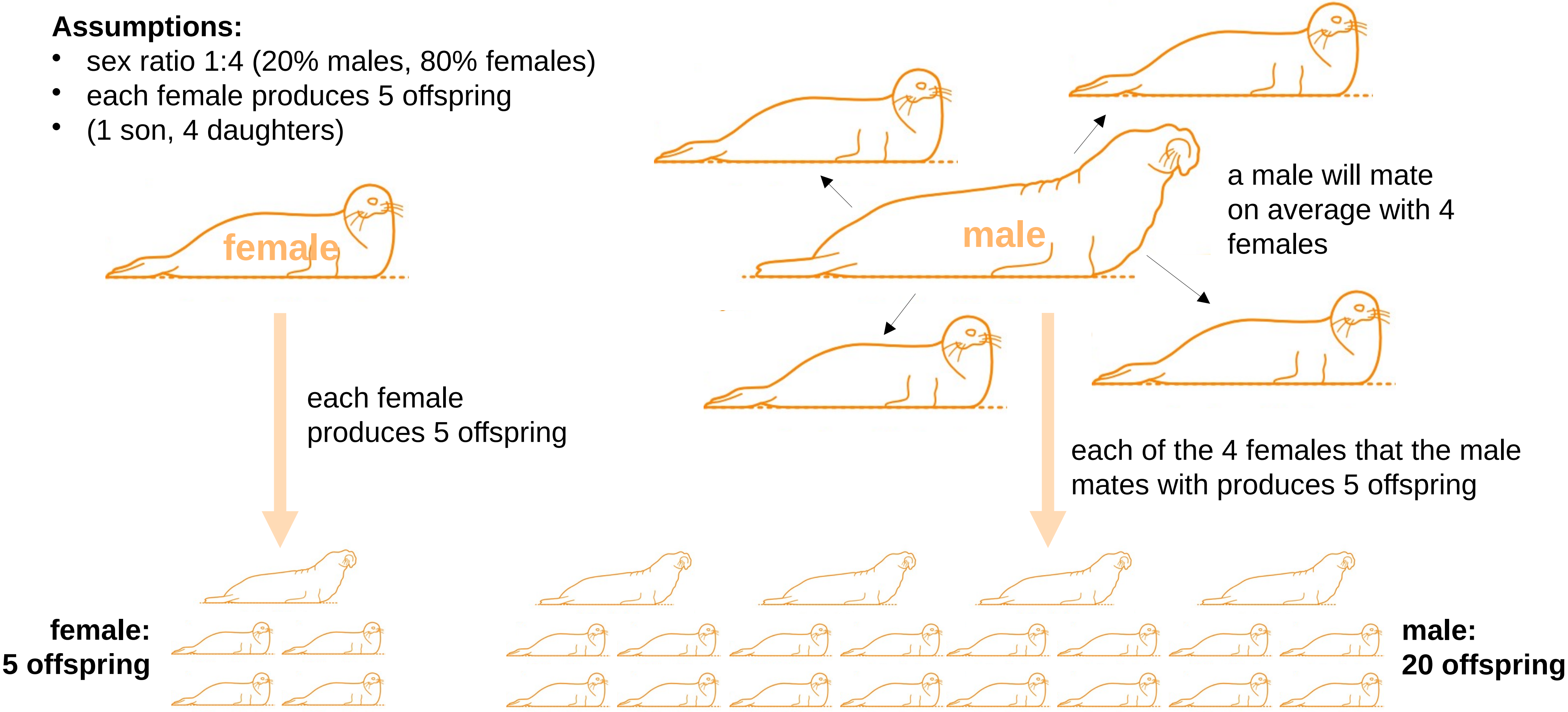
Sea elephants

- males control large harems of females
- only a minority of males mates at all (most males die without ever mating successfully)
- it seems “ineffective” that, nonetheless, males and females are produced in equal proportions
- imagine a population where females produce 20% males and 80% females, so that all males enjoy reproductive success and none are “wasted”

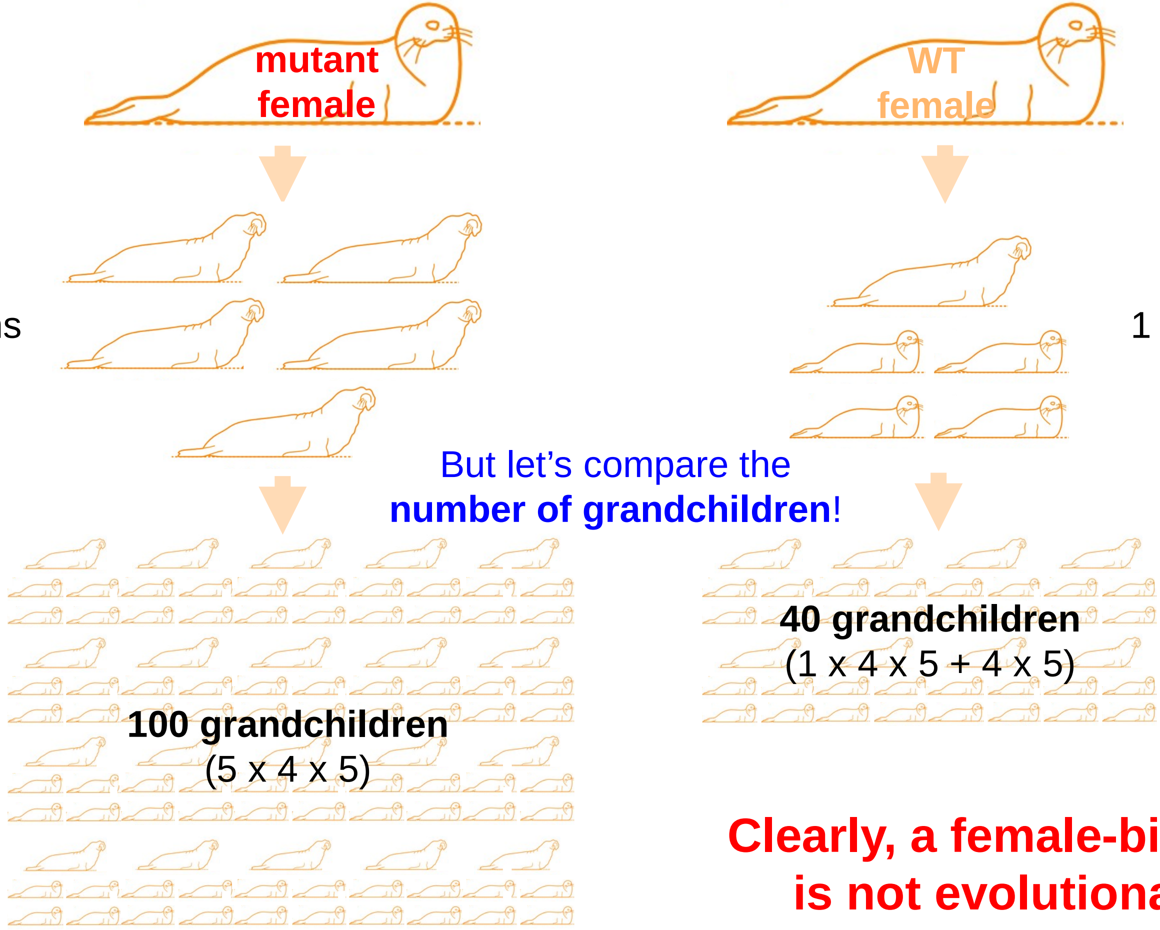
Why is this strategy not found in nature?
Why is it not evolutionarily stable?

Comparing the number of offspring of females vs males

Assumptions:
- sex ratio 1:4 (20% males, 80% females)
- each female produces 5 offspring
- (1 son, 4 daughters)



Now imagine a mutant female that produces only sons!



5 sons

1 son, 4 daughters

But let’s compare the
number of grandchildren!

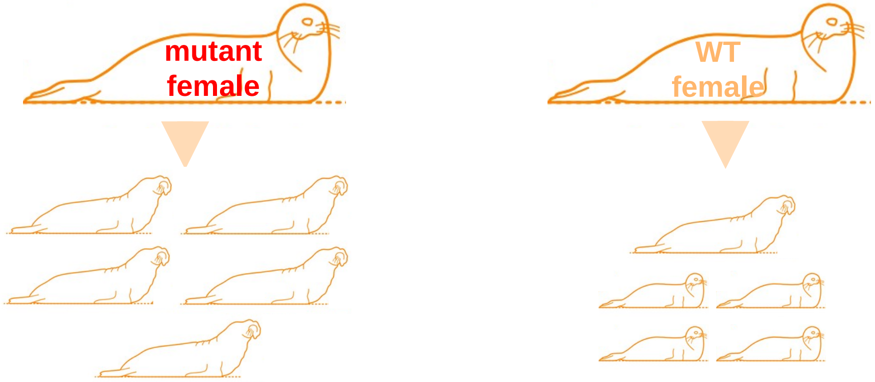
each son mates
with 4 females who
each produce
5 offspring

1 son who mates with
4 females who each
produce 5 offspring
and 4 daughters who
each produce 5 offspring

**Clearly, a female-biased sex ratio
is not evolutionarily stable!**

Two very important points

1 **How to define fitness?**

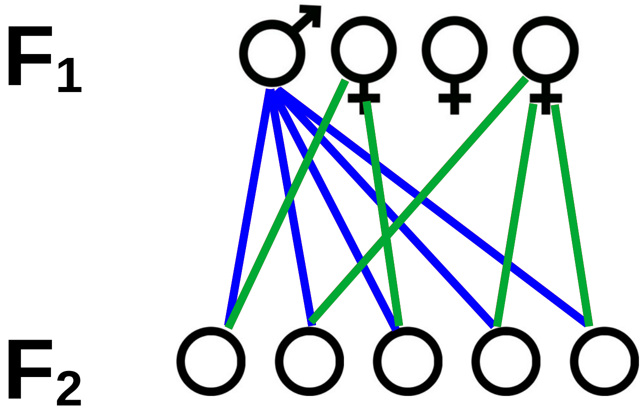


~~**Fitness:** number of surviving offspring~~

Fitness here: number of grandchildren!

2 **The assumption that every individual has**
exactly one father and one mother

Every individual has exactly one father and one mother



Every F_2 individual has
an F_1 father and an F_1 mother

Total number of offspring from F_1 males = **Total number of offspring from F_1 females**

Arriving at an ultimate explanation for a balanced sex ratio

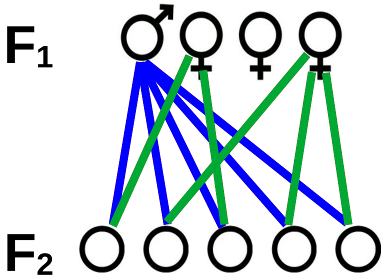
Total number of offspring from F_1 males	=	Total number of offspring from F_1 females

s_m = proportion of F_1 males

w_m = average fitness of F_1 male

s_f = proportion of F_1 females

w_f = average fitness of F_1 female



(1/4 · 5) $\boxed{s_m \cdot w_m = s_f \cdot w_f}$ (3/4 · 5/3)

$w_m = 3w_f$ $\boxed{w_m = s_f / s_m \cdot w_f}$

The rarer sex always has a higher fitness

It always pays to produce the rarer sex

This explanation has become known as Fisher’s principle

In fact, Fisher’s principle is not (only) Fisher’s principle

We might just call it the “rarer-sex effect”:

Individuals producing the rarer sex have an evolutionary advantage, which thereby increases the frequency of the sex they produce.

The rarer-sex effect

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The rarer-sex effect

Andy Gardner (2023)

School of Biology, University of St Andrews, St Andrews KY16 9TH, UK

Charles Darwin (1871)	Carl Düsing (1883)	Corrado Gini (1908)	John A. Cobb (1914)	Ronald A. Fisher (1930)	William D. Hamilton (1964-70)

It was Düsing who made the two important points

1. **Counting grandchildren as a measure of reproductive success**

2. **Assuming that every Individual has exactly one father and one mother**

Die

Regulierung

des

Geschlechtsverhältnisses

bei der

Vermehrung der Menschen, Tiere und Pflanzen.

Von

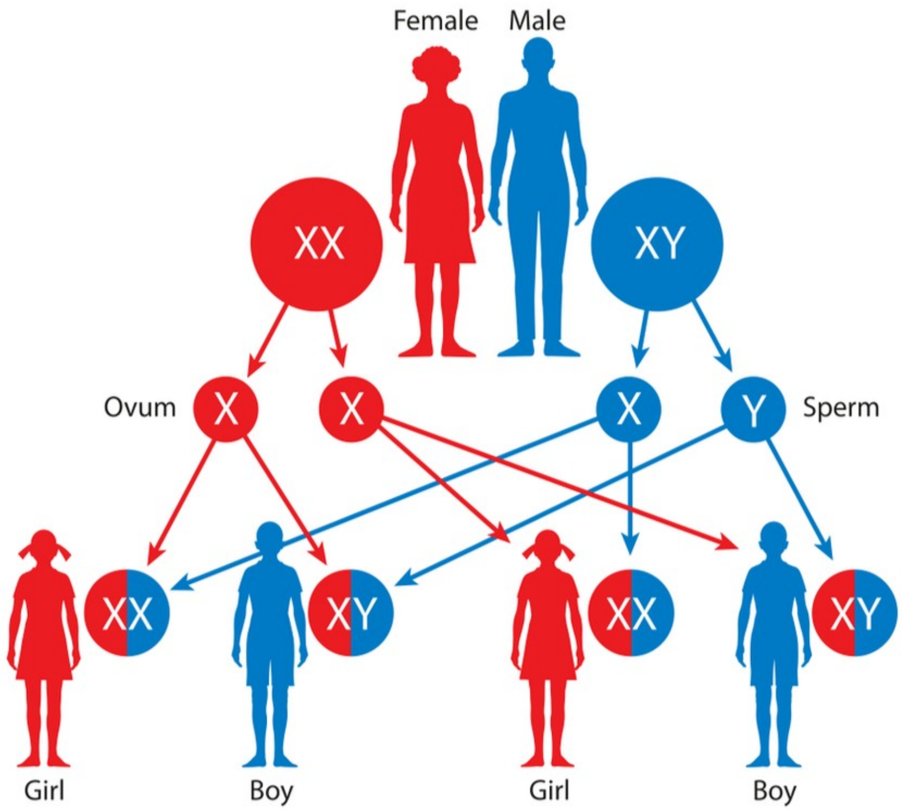
Carl Düsing,
Dr. phil.

Mit einer Vorrede
von
Dr. W. Preyer,
o. ö. Professor der Physiologie und Director des physiologischen Institutes der Universität Jena.

(Separatabdruck aus der Jen. Zeitschrift für Naturwissenschaft. Bd. XVII. N. F. X. Bd.)

Jena,
Verlag von Gustav Fischer
1884.

Having one father and one mother – isn’t that obvious?



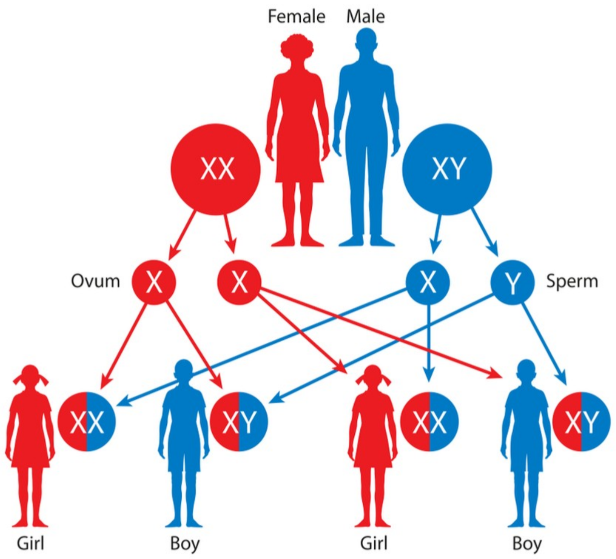
Well, in humans it is, but there are many other possibilities!

To see this, we need to go back to proximate explanations.

Mechanisms of sex determination

XY sex determination

Sex chromosomes: X, Y



Genotypes

- **XX (female)**
- **XY (male)**

Examples

Humans, most other mammals

ZW sex determination

Sex chromosomes: Z, W

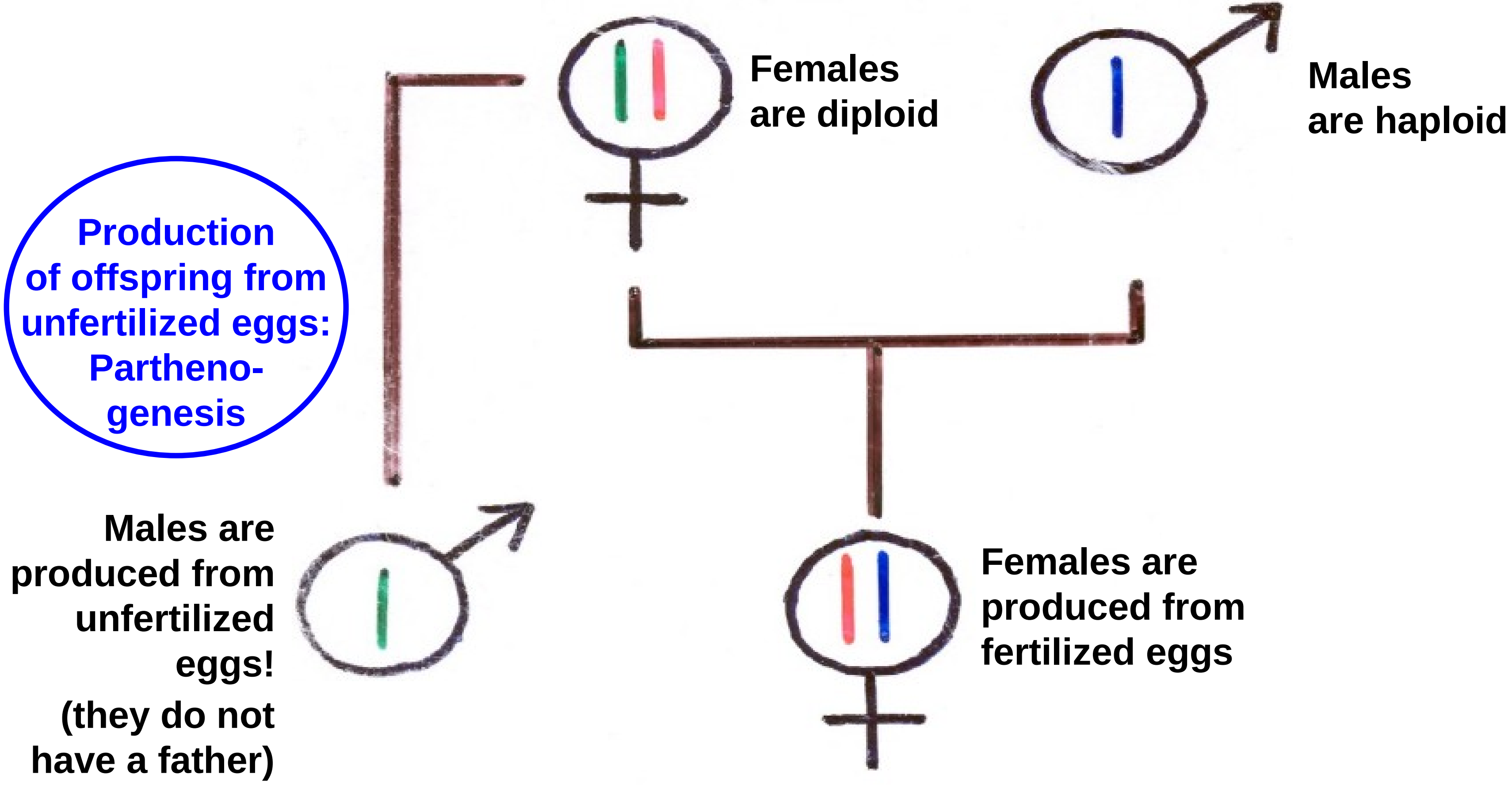
Genotypes

- **ZZ (male)**
- **ZW (female)**

Examples

Birds, some insects

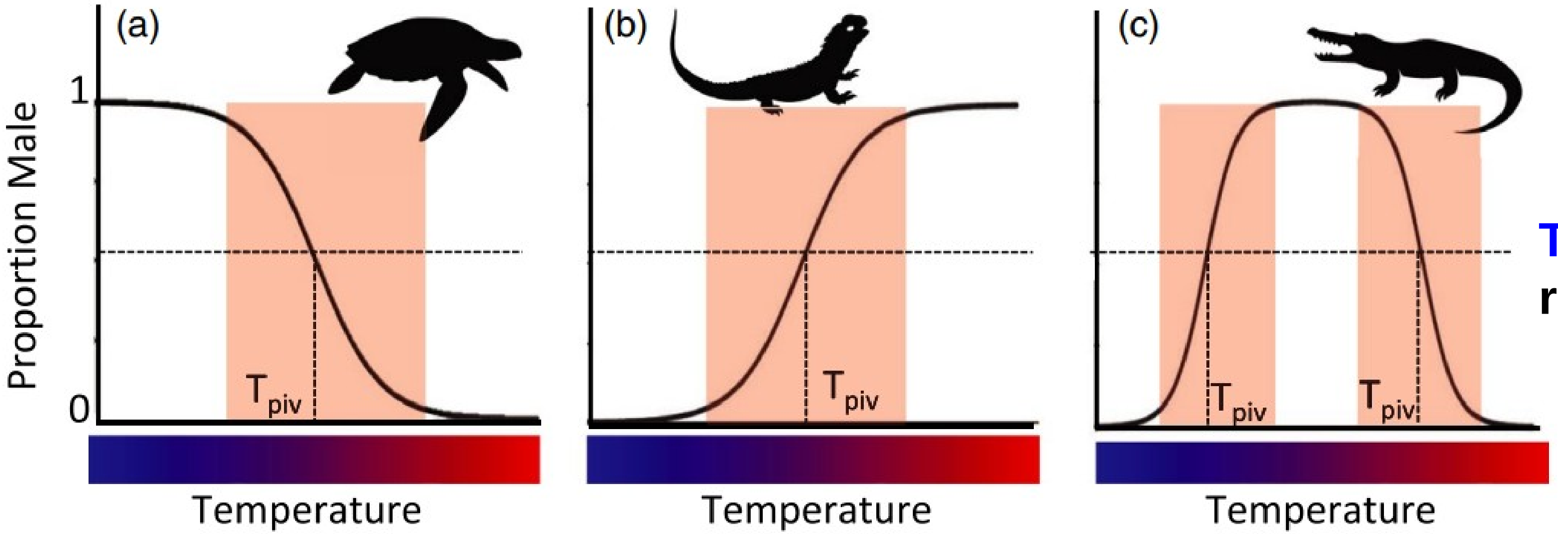
Haplodiploidy



Examples of sex determination though haplodiploidy

Ants, bees, wasps, some mites

Environmental sex determination



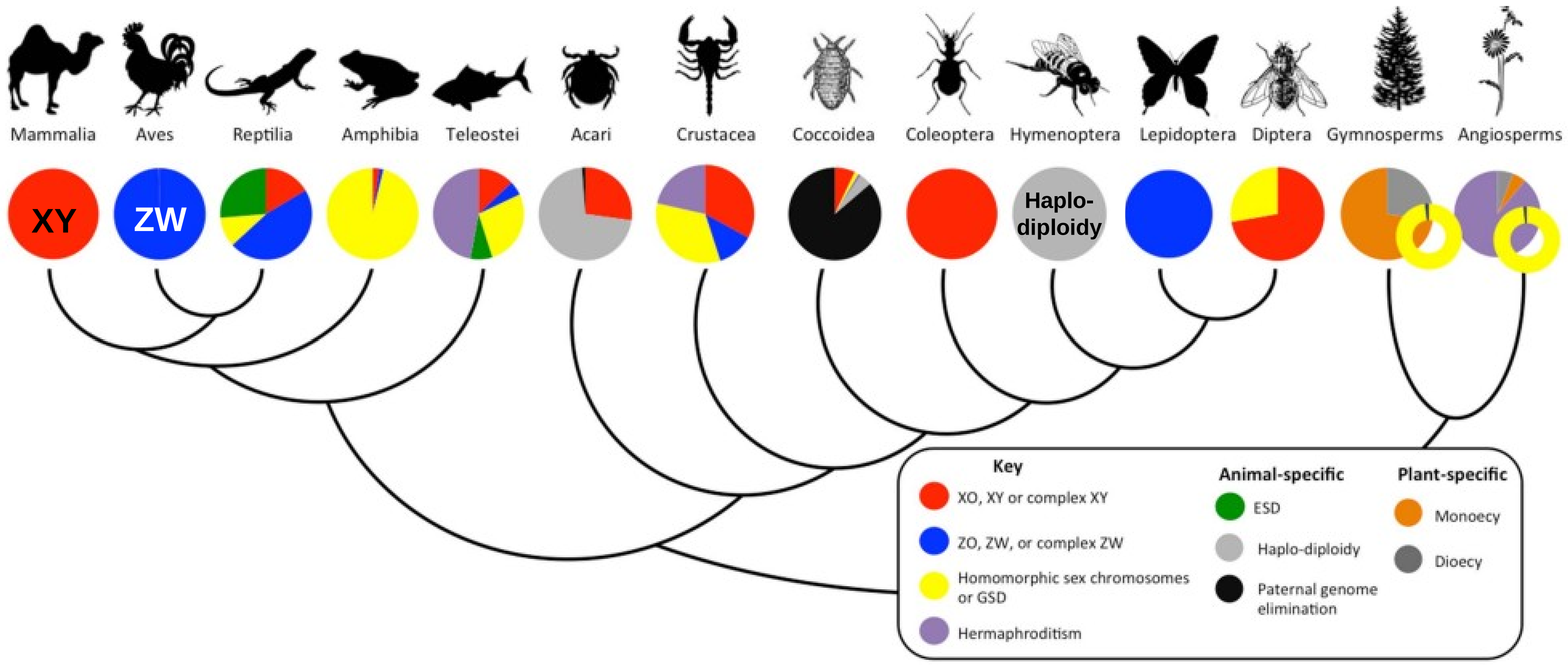
Lockley & Eizaguirre, 2021

Temperature: **reptiles, fish**

Photoperiod: ***Gammarus duebeni***

Location: **green spoonworm**

Diversity of sex determination systems



Bachtrog et al., 2014

Fisher's principle explains why "ordinary" sex ratios are 1:1

Extraordinary Sex Ratios

A sex-ratio theory for sex linkage and inbreeding has new implications in cytogenetics and entomology.

W. D. Hamilton

"This article is concerned with situations where certain underlying assumptions of Fisher's argument do not hold." (Hamilton, 1967)

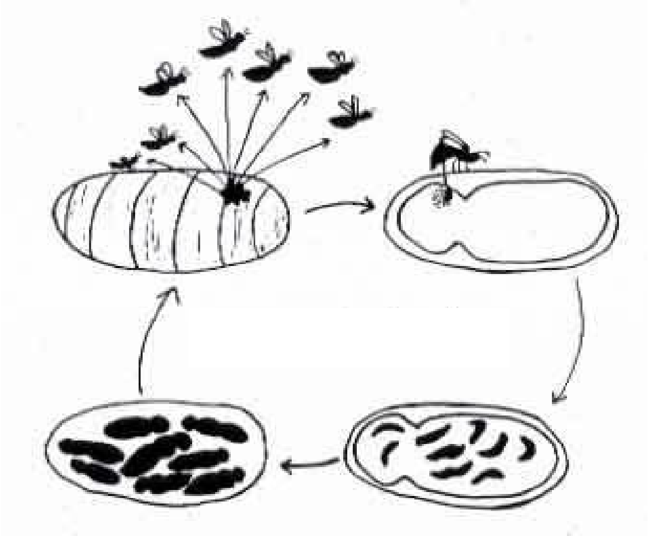
One of these assumptions is **panmixia / random mating**: each male of a population has equal chance of mating with any female in the population, and vice versa

"population-wide competition for mates"

What happens if this assumption is violated (i.e., **if mating is non-random**)?

"local competition for mates"

Local mate competition (LMC)



- a small number of females (*n*) lay eggs in one host
- offspring mate among themselves (**sibmating**)
  → sons compete with each other for mates (**local mate competition**)
- only the females disperse to found new subpopulations

In such a situation, Hamilton's (1967) theory of LMC predicts female-biased sex ratios

Sibmating as a prerequisite for female-biased sex ratios

Under LMC, the unbeatable sex ratio is (n – 1)/2n (Hamilton, 1967)

***n* = 1:**
unbeatable sex ratio approaches zero

sibmating extremely high

strong LMC

a single male is enough to fertilize all his sisters

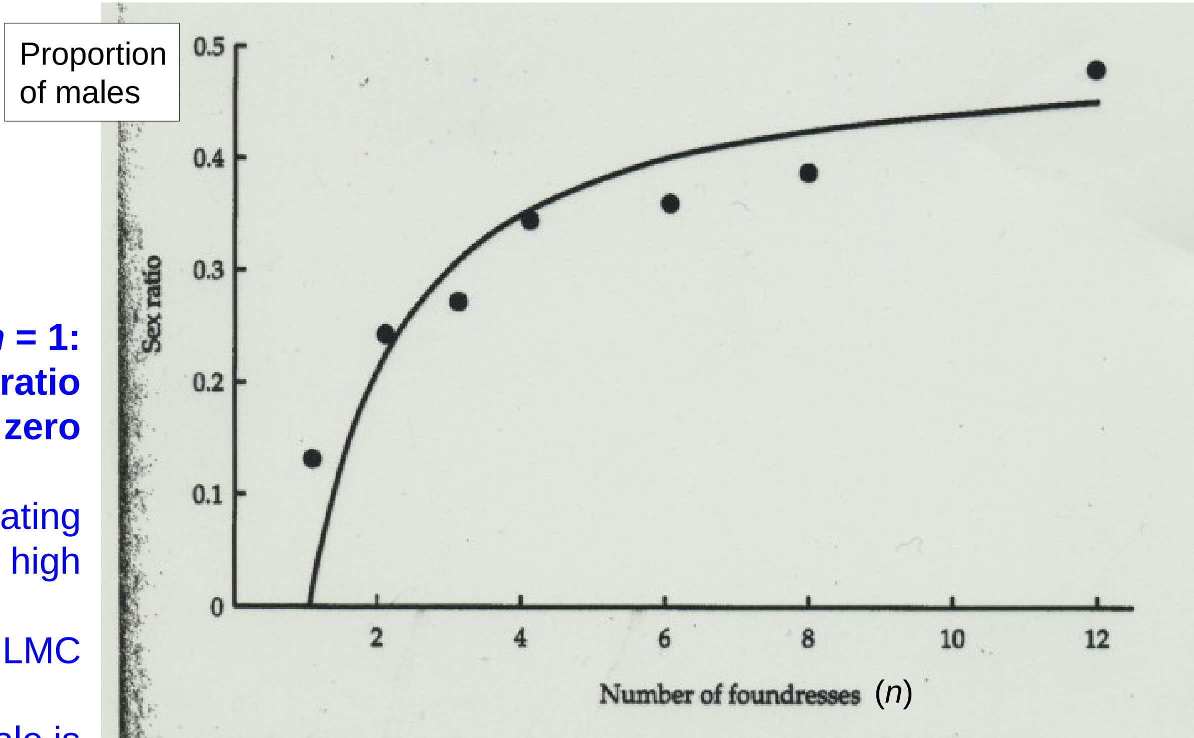


Figure 10.3. Sex ratios of a parasitic wasp from patches containing different numbers of foundresses in a laboratory experiment. The curve shows the theoretical prediction. After Werren, 1988.

***n* high:**
unbeatable sex ratio approaches 0.5

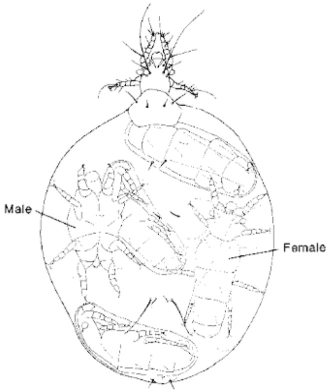
sibmating low (more or less random mating)

no LMC

Fisherian 1:1 sex ratio

An extreme case of sibmating

***Acarophenax tribolii*, a viviparous mite**



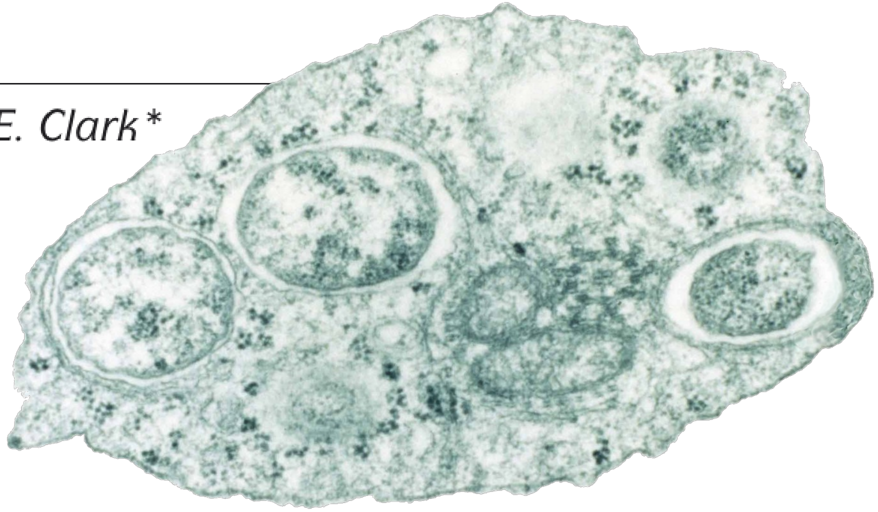
"One example in particular totally blew my mind: *Acarophenax tribolii*. These mites usually occur in broods biased 15-1 in favor of females, and they are quite naughty. The single male will mate with and fertilize all of his sisters. Hang on. Before you get too excited, all this happens in full view of the mother. Is your jaw slack? This will make your jaw hit the floor. All this takes place *in* the mother's womb! That's right. The enormously swollen, gravid *A. tribolii* female has ~16 full-grown adult mites having a great, big, incestuous orgy in her womb! Eventually these ungrateful offspring burst her open, killing her. Actually not all of them. The poor male completes his life cycle and dies before he is born. At least he isn't a virgin."

(John Dennehy; https://evilutionarybiologist.blogspot.com/2007/10/extraordinary-sex-ratios.html)

Another cause of female-biased sex ratios

Wolbachia: master manipulators of invertebrate biology

John H. Werren, Laura Baldo‡ and Michael E. Clark**



intracellular bacteria
infecting many insect species

How are *Wolbachia* bacteria transmitted?

Transmission of *Wolbachia*

Transmission of *Wolbachia* is maternal

From mother to offspring via egg cells

Transmission of *Wolbachia* is maternal

For *Wolbachia*,
being in a male host is a dead end
(a property shared with mitochondria)

***Wolbachia* are under selection**
to increase the proportion of infected females

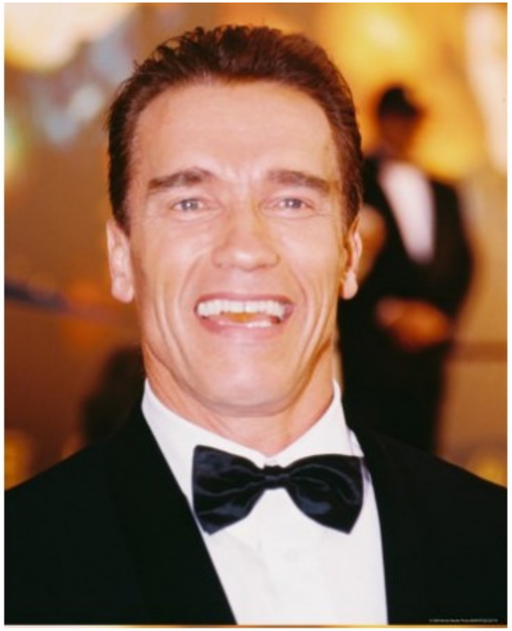
What strategies could *Wolbachia* use to achieve this?

Feminisation of males

But not like this!

Feminised males would need to produce eggs!

Sounds like science fiction!



Or maybe not?

Strategies of *Wolbachia* to induce female-biased sex ratios

- **Feminisation**
  (transformation of males into females)
- **Induction of parthenogenesis**
  (unfertilized eggs develop into females)
- **Male killing**

Strategy 1: Feminisation

***Wolbachia*-infected males develop into females**

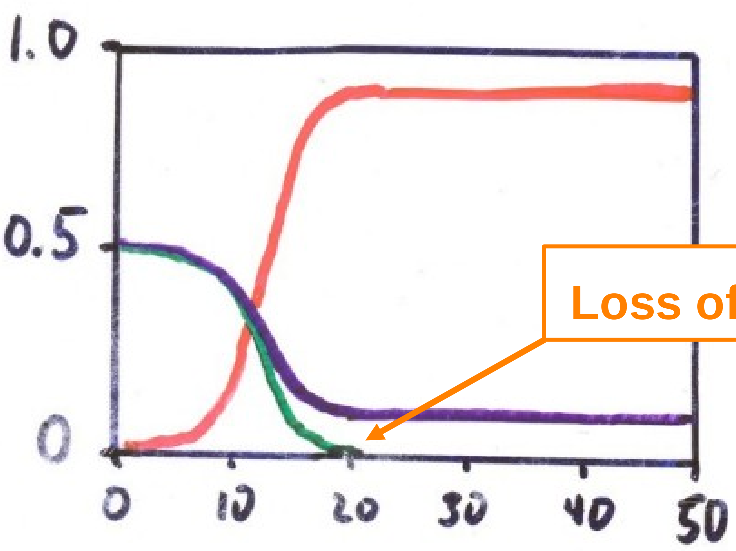
Sex determination in *Armadillidium vulgare*

Chromosomal sex	Functional sex
ZW	**Female**
ZZ	**Male**
ZZ + *Wolbachia*	**Female**

Model of an infected population

(assuming 90% transmission efficiency of *Wolbachia*)

Proportion of individuals



Generations

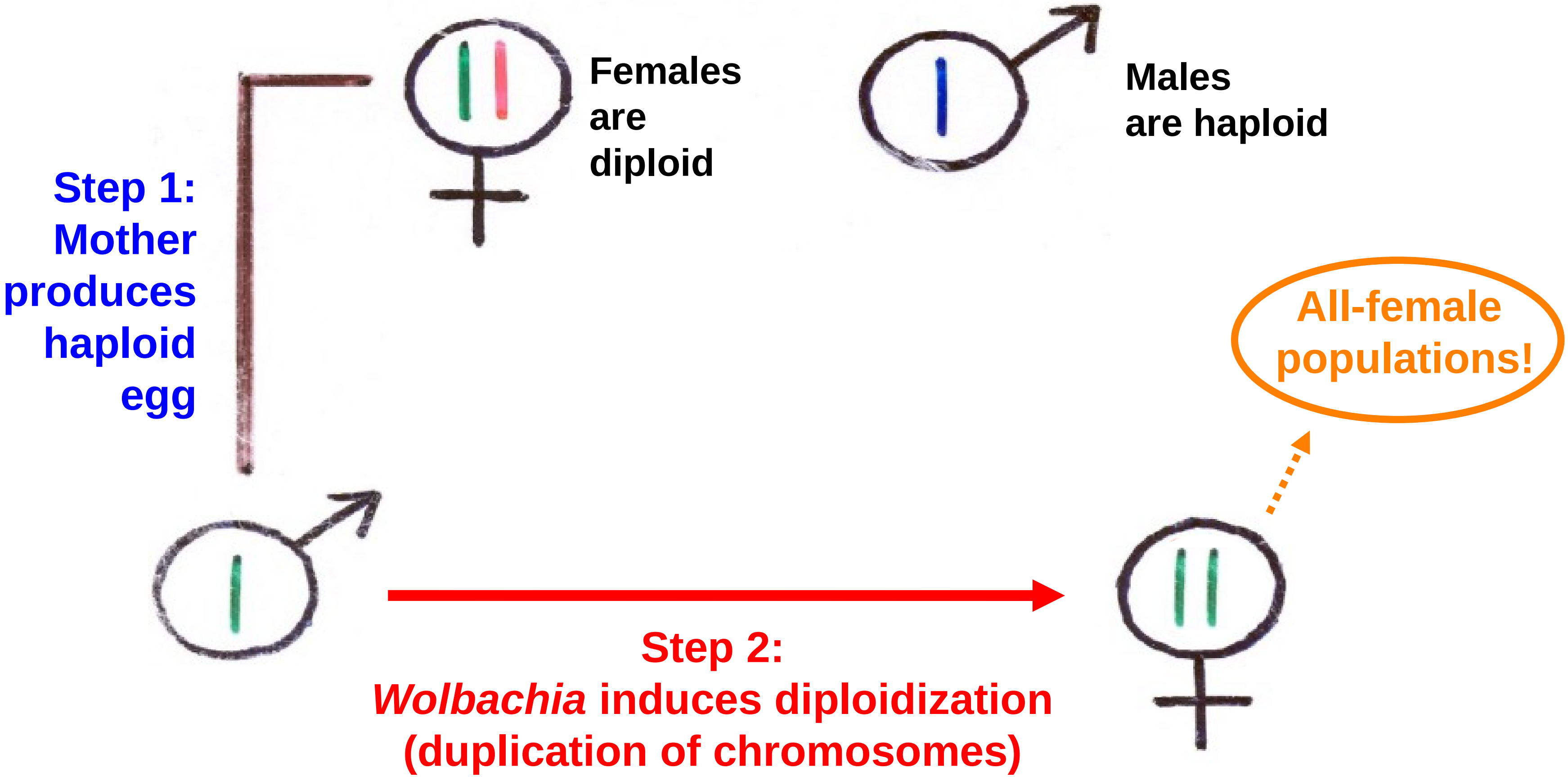
ZZ + *Wolbachia* (feminised males/ transsexual females)

Loss of W chromosome!

ZZ (genetic males)

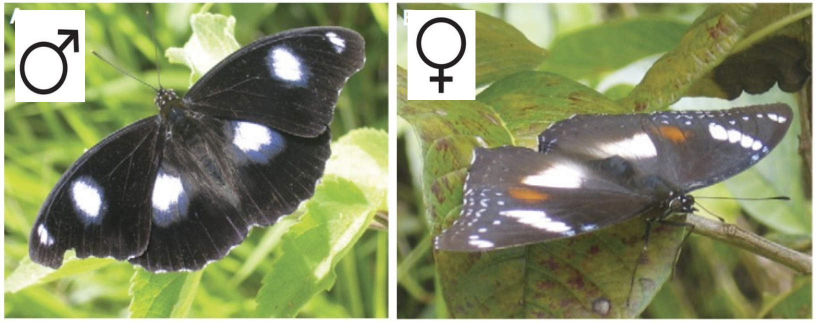
ZW (genetic females)

Strategy 2: Induction of parthenogenesis (only in haplodiploids)



Strategy 2: Male killing

In the 1920s, butterflies collectors in Fiji noticed an extreme sex ratio in *Hypolimnas bolina*:



1 male per 100 females!

What was going on there?

Wolbachia-induced male killing

***Wolbachia* kill male eggs of infected females,**
so that sisters of killed males obtain a fitness benefit!

- **In 2001, the extreme sex-ratio bias (1:99) was still present**
- **this persistence of male killing is difficult to understand: a mutant male that escapes male killing would initially have a strong selective advantage (because of the rarer-sex effect)**
- **very strong selection on hosts to suppress male-killing (evolution of resistance)**

In 2006, the population sex ratio had changed from 1:99 to 50:50

→ evolution of resistance within only 10 generations!

The Strange Case of the Butterfly and the Male-Murdering Microbe

A battle between an insect and a microbe led to one of the fastest evolutionary changes ever observed.

By Ed Yong

The Atlantic

Ordinary and extraordinary sex ratios

- ordinary 50:50 (“Fisherian”) sex ratios can be explained by the rarer-sex effect
- extraordinary (female-biased) sex ratios can be explained by:
  - local mate competition under inbreeding (Hamilton, 1967)
  - bacteria that selfishly manipulate host sex ratio, such as *Wolbachia*

A glossary of some biological terms from Hamilton (1967)

Arrhenotoky	a form of parthenogenesis where unfertilized eggs develop into males (adjective: arrhenotokous or arrhenogenous)
Biofacies	co-occurrence of characteristic biological features
Drive	e.g. **meiotic drive** (gametic drive): when particular alleles are transmitted to more than half (or sometimes even all) of the offspring (as opposed to the Mendelian 50:50 ratio)
Dysgenic	genetically detrimental
Heteropycnosis	if some chromatin is more/less intensely stained than the “standard” chromatin
Impaternate	produced parthenogenetically by a female without fertilization (“without a father”)
Inbreeding	mating of closely related individuals (opposite: outbreeding)
Male haploidy	= **haplodiploidy**: males are haploid (and produced via arrhenotoky), females are diploid
Monogamy	a mating system in which one male and one female mate exclusively with each other (adjective: monogamous)
Monogyny	a mating system in which a male only mates with one female (adjective: monogynous)
Outbreeding / outcrossing	mating of unrelated (or not closely related) individuals (opposite: inbreeding)
Panmixis / panmixia	random mating (adjective: panmictic)
Parthenogenesis	A form of asexual reproduction where embryos develop from unfertilized eggs
Polygamy	A mating system that is non-monogamous (adjective: polygamous)
Polygyny	a mating system in which one male mates with multiple females (adjective: polygynous)
Sibmating	mating between siblings (a form of inbreeding)
Spanandry	when there are much more females than males (adjective: spanandrous)
Thelytoky	A form of parthenogenesis where unfertilized eggs develop into females (adjective: thelytokous or thelygenous)